database_sharding_strategies (Initial Draft)

Domain: Software Architecture | Document ID: database_sharding_strategies_v1
Specification Standard: IEEE Capstone Evaluation Corpus

Executive Overview:

Horizontal database partitioning, consistent hashing rings, virtual nodes,
and cross-shard querying trade-offs.

Methodological Verification & System Lineage:

This specification is catalogued in IntentCloud for intent-aware hybrid retrieval.
Text chunks are indexed into 384-dimensional dense semantic vectors and
1,000,003-dimensional sparse feature-hashed token projections.
Candidate pools are unified using scale-invariant Reciprocal Rank Fusion (k=60).
Explainable citation attribution is extracted using cross-encoder cross-attention.